Smart Grid Decision Load Agent

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Abstract

- This project presents a **Smart Grid Load Decision Agent** that allocates available electrical power to different demand sectors (e.g., industrial, residential, hospital) based on priority levels. The agent implements a deterministic, rule-based decision strategy that sorts sectors by priority (highest first) and allocates power up to their requested demand, respecting the total available capacity. A complete trace log records each allocation step, enabling full explainability. The system is designed to handle emergency scenarios where available power is insufficient, ensuring critical sectors receive power first.

Goals

- **Primary Goal:** Maximise the supply of power to high-priority sectors under a limited total power budget.
- **Secondary Goal:** Provide a transparent, step-by-step reasoning trace for each allocation decision.
- **Constraint Satisfaction:** Never allocate more power than the available total.
- **Fairness & Efficiency:** Avoid waste by allocating exactly the requested demand whenever possible, and use remaining power fairly among lower-priority sectors.

Problem Statement

- In a smart grid environment, the central controller must decide how to distribute limited electrical power among multiple demand sectors (e.g., hospital, industrial, residential). Each sector has:
- A **power demand** (kW) that it would ideally consume.
- A **priority** (1 = lowest, 5 = highest) indicating its criticality.
- The total available power is fixed and may be less than the sum of all demands. The agent must allocate power to sectors in a way that:
- Never exceeds the total available power.
- Respects priority (higher priority sectors get their full demand first).
- If power is insufficient, lower-priority sectors receive whatever remains, possibly less than their demand.
- Provide a clear, auditable trace of all decisions

System Environment

- **Agent:** SmartGridAgent (decision-making entity).
- **Environment:** Static, deterministic, discrete time (single allocation round). The environment consists of a set of sectors, each with known demand and priority, and a single numerical value representing total available power.
- **Actuators:** The agent sets the allocated attribute for each sector.
- **Sensors:** The agent reads sector demands, priorities, and the available power value (input by user).
- **Interface:** Command-line input/output; the user provides available power, number of sectors, and each sector's name, demand, and priority.

Observations

- **Priority Sorting Works:** The agent correctly sorts sectors by descending priority before allocation. For example, a hospital (priority 5) will be served before residential (priority 3).
- **Constraint Handling:** When remaining power is less than a sector's demand, the agent activates the constraint, allocates the remaining power to that sector, and stops further allocation.
- **Trace Clarity:** Each allocation step is logged, including when a constraint is triggered. This makes the decision process fully explainable.
- **Edge Cases:**
  - If two sectors have the same priority, the one listed first (after sorting) receives power first – this can be improved by tie-breaking rules.
  - If available power is zero, all sectors receive zero allocation.
- **Limitation:** The agent does not consider future time steps or variable renewable generation; it is a one-shot allocator.

Conclusion

- The Smart Grid Load Decision Agent successfully implements a priority-based power allocation strategy that is simple, efficient, and transparent. It satisfies all stated goals and constraints. The trace log provides valuable insight for system operators and can be extended to include more sophisticated reasoning, such as probabilistic demand forecasting, search-based optimisation (UCS/A*), or constraint satisfaction (CSP) for multi-period scheduling. The current version serves as a solid baseline for more advanced smart grid decision agents that incorporate uncertainty, learning, and multi-agent coordination.